

Agent Architecture Patterns

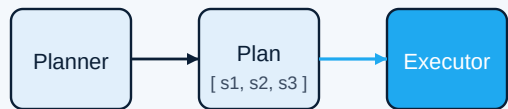
FOUR PATTERNS · CONTROL FLOW · WHEN TO USE EACH

Control flow
Data flow

PATTERN 01

Planner–Executor

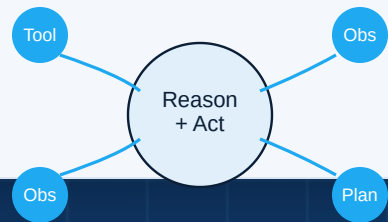
Static plan decomposed once; executor realizes steps; deterministic & auditable.



PATTERN 02

ReAct

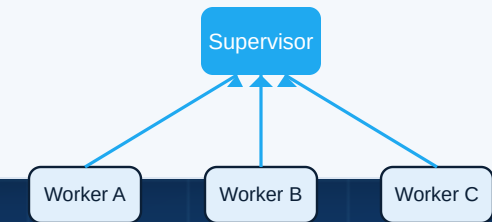
Interleaved Reason → Act → Observe loop; adapts mid-flight; tool-grounded.



PATTERN 03

Supervisor

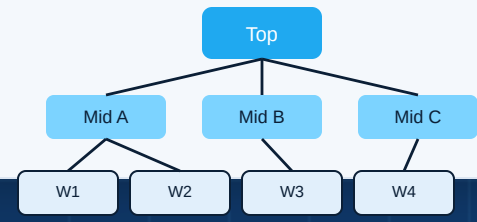
Central dispatcher routes tasks to specialist workers; one routing brain.



PATTERN 04

Hierarchical

Layered controllers; top-level strategy delegates to sub-supervisors.

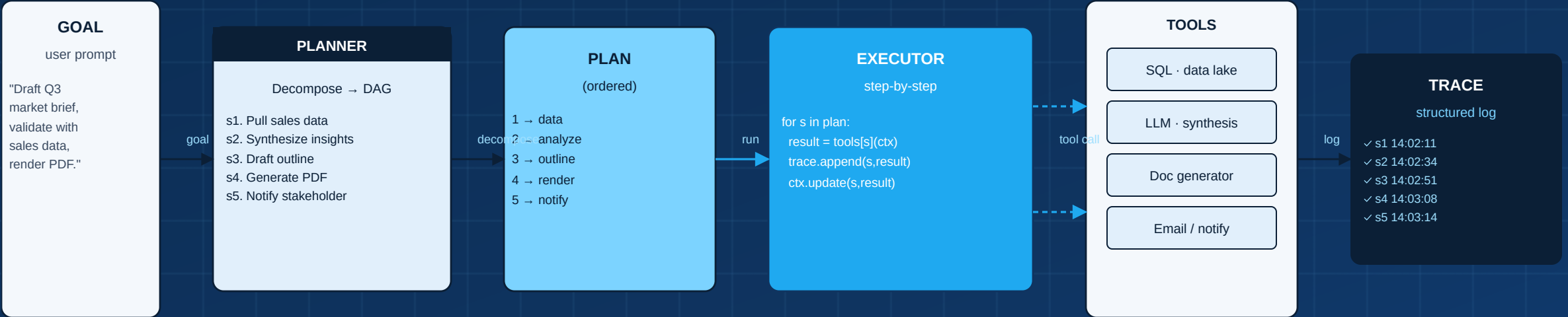


PATTERN SELECTION MATRIX

PATTERN	PLANNING MODE	BEST FOR	TRADEOFF
Planner–Executor	Up-front, static	Multi-step pipelines, SOPs, audit-bound workflows	Rigid to mid-flight change
ReAct	Per-step, dynamic	Unknown envs, exploration, tool-rich tasks	Higher latency / cost
Supervisor	Delegated, role-based	Multi-domain specialists, parallel workstreams	Single router is a bottleneck
Hierarchical	Tiered strategy	Enterprise-scale orchestration, recursion	Complex governance design

Planner-Executor

PLAN ONCE · EXECUTE STEP-BY-STEP · LOG EVERY TRANSITION



STRENGTHS

- Predictable cost & latency
- Full execution trace
- Parallel branches via DAG

USE WHEN

- Workflows are well understood
- Compliance / audit required
- Steps have stable tools

FAILURE MODES

- Bad plan → cascading errors | No mid-flight rewrite | Tool failures need explicit retry policy | Plan drift if prompt changes

ReAct · Supervisor · Hierarchical

DYNAMIC PLAN · ROLE ROUTING · TIERED CONTROL

PATTERN 02 · REACT

Reason → Act → Observe loop. Each step reassesses situation.



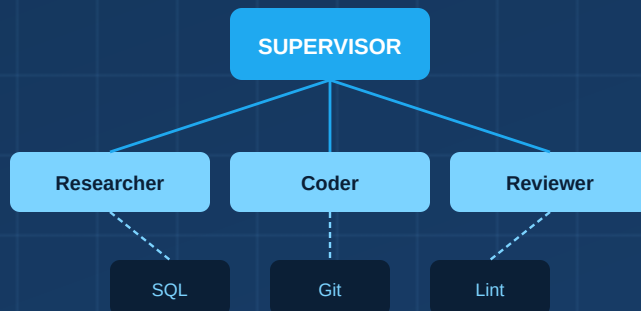
SIGNATURE

```
Thought: I need user age.  
Action: get_user(id)  
Obs: age = 34  
Thought: Now recommend...
```

- Adapts to new info
- Tool-grounded actions
- Costly per turn

PATTERN 03 · SUPERVISOR

Central router dispatches to specialist workers; aggregates results.



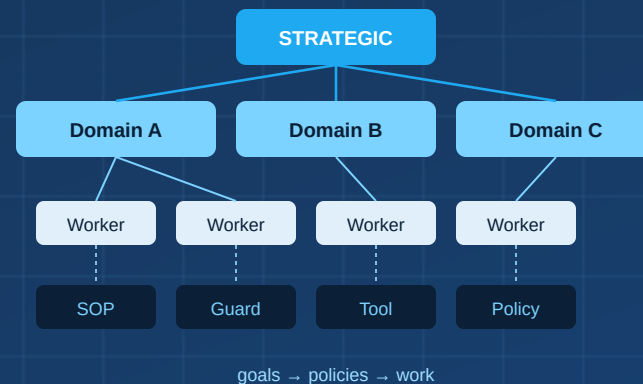
ROUTING LOGIC

```
route = llm.decide(  
    task,  
    workers=[  
        "researcher",  
        "coder",  
        "reviewer"]  
)
```

- Single point of decision
- Parallel workers possible
- Router is critical path

PATTERN 04 · HIERARCHICAL

Layered supervisors; top tier sets strategy, mid tiers execute.



- Scales to enterprise
- Governance per layer
- Cross-tier messaging

LangGraph for Multi-Agent Orchestration

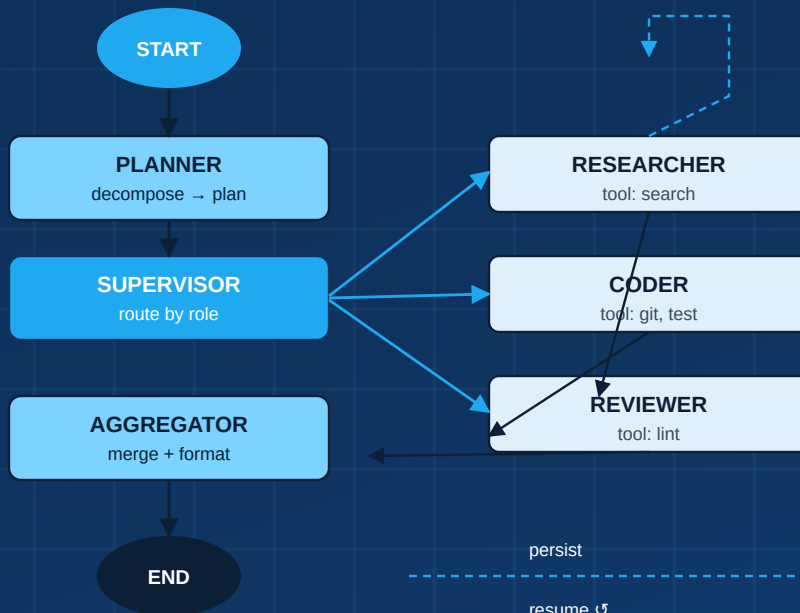
GRAPH STATE · NODES · EDGES · CYCLES · CHECKPOINTS

STATE CHANNEL (TypedDict)

```
messages: List[BaseMessage]
next: str
plan: List[Step]
results: Dict[str, Any]
tool_calls: List[ToolCall]
memory: ConversationBuffer
errors: List[Exception]
budget: {tokens, usd}
checkpoint_id: UUID
```

SHARED · IMMUTABLE BETWEEN NODES · REDUCER-MERGED

NODES (workers)



CHECKPOINT

SQLite / Postgres

thread_1 · t=42

thread_2 · t=12

thread_3 · t=7

```
store.save(state)
store.load(checkpoint)
resume / replay
```

TOOL REGISTRY

```
search · sql · git
lint · notify · file
mcp://tools/*
```

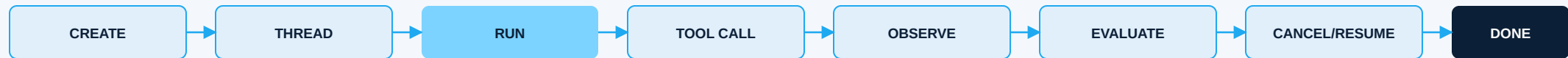
```
graph = StateGraph(State)
graph.add_node("planner", planner_fn); graph.add_node("supervisor", sup_fn); graph.add_node("workers", worker_fn)
graph.add_conditional_edges("supervisor", route, {"researcher": "workers", "coder": "workers", "done": "aggregator"})
memory = MemorySaver(); app = graph.compile(checkpointer=memory)
```

Core Features · cyclic graphs · typed reducers
human-in-the-loop · time-travel · parallel branches
per-node retry · streaming

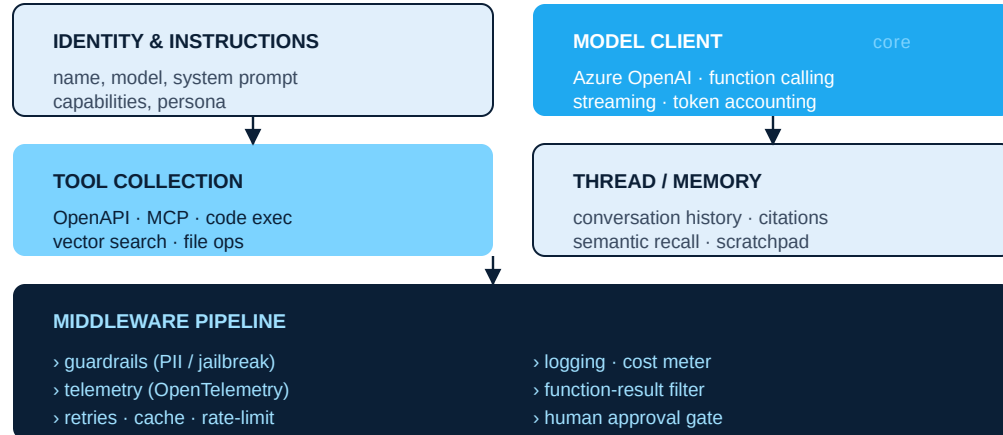
Microsoft Agent Framework

AGENT · THREAD · RUN · TOOL · MIDDLEWARE

RUN LIFECYCLE



AGENT ANATOMY



WORKFLOW ORCHESTRATION

> GroupChat

manager selects next speaker by policy
round-robin · dynamic · hand-off

> HANDOFFS

agent A → agent B with payload
typed contracts · ack/nack

> MAGENTIC

orchestrator + ledger
plan / progress / fact-check

> CONCURRENT FAN-OUT · GUARDED FAN-IN

run n agents in parallel with shared context
aggregate via reducer; abort on guardrail fail

```
agent = ChatAgent(model="gpt-4o", instructions="...", tools=[tool1, tool2])
thread = agent.create_thread(); async for evt in agent.run_stream(thread, msg): ...
wf = GroupChatOrchestrator([researcher, coder, reviewer], policy=handoff())
```

Strengths · Azure-native · typed contracts
built-in Magentic ledger · OpenTelemetry hooks
Entra ID auth · APIM-fronted endpoints

Choosing an Orchestration Framework

STATE MODEL · CONTROL MODEL · HOSTING · OBSERVABILITY

LANGGRAPH

GRAPH · CYCLES · IN-PROCESS

CONTROL MODEL

StateGraph + reducers
explicit nodes & edges
loops, branches, fan-out

STATE

```
class State(TypedDict):  
    messages: Annotated[...]  
    plan: List[Step]  
    results: Dict[...]
```

HOSTING

Python service

Checkpointing (SQL)

MCP tools

BEST FOR

Custom control logic · loops · time-travel · open-source

MICROSOFT AGENT FRAMEWORK

AGENT · THREAD · MANAGED

CONTROL MODEL

GroupChat, Magentic
handoffs & concurrent
policy-driven, declarative

STATE

```
thread = agent.create_thread()  
async for evt in run_stream():  
    store.append(evt)  
ledger.checkpoint()
```

HOSTING

Azure AI Foundry

Entra ID + APIM

MCP tools

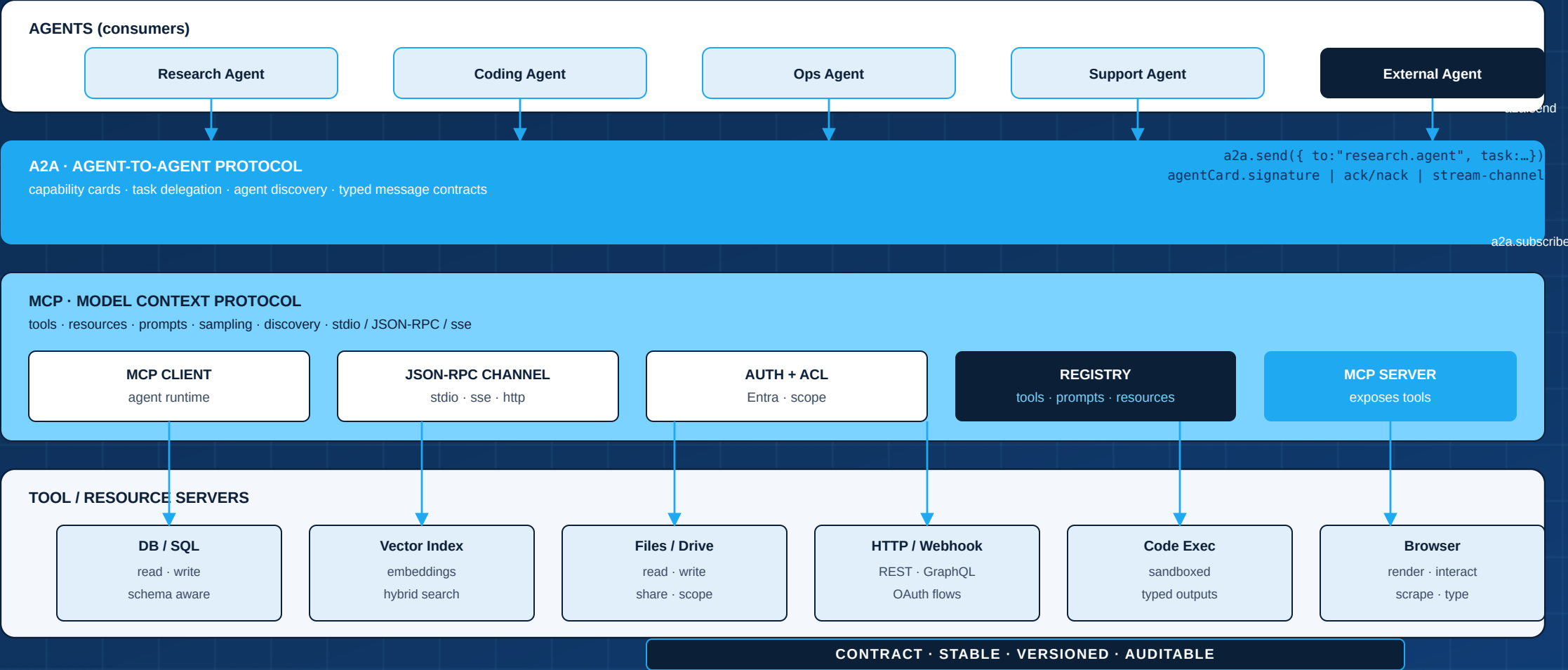
BEST FOR

Enterprise governance · Azure-native · managed runtime

DECISION DRIVER	PICK LANGGRAPH	PICK MICROSOFT AGENT FRAMEWORK
Cycles & time-travel	Native; replay from any checkpoint	Limited; rely on managed ledger
Azure integration	DIY wiring of Entra / APIM	First-class; Foundry hosting
Custom state reducers	First-class TypedDict reducers	Schema-bound; less ergonomic
Governance	App-level middleware	Built-in ledger & OpenTelemetry

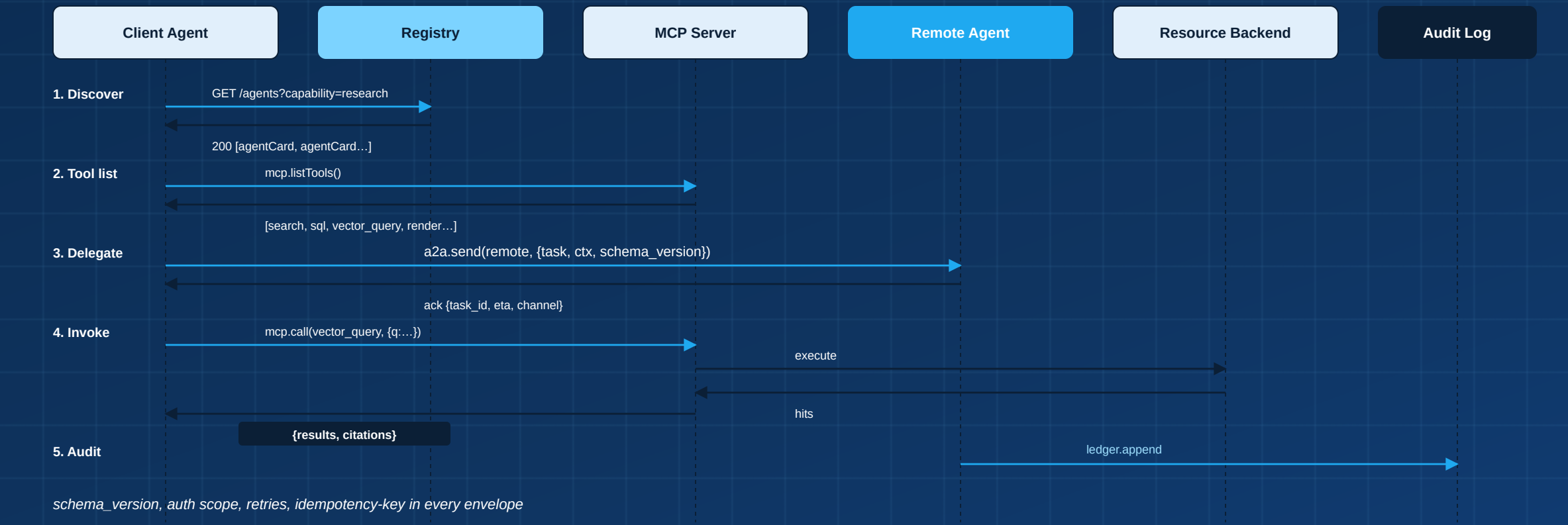
MCP & A2A as Integration Contracts

PROTOCOL BOUNDARIES · TOOL EXPOSURE · AGENT INTEROP



Discovery · Registration · Message Exchange

AGENT CARDS · TOOL REGISTRIES · TYPED CONTRACT FLOWS



DISCOVERY & REGISTRATION

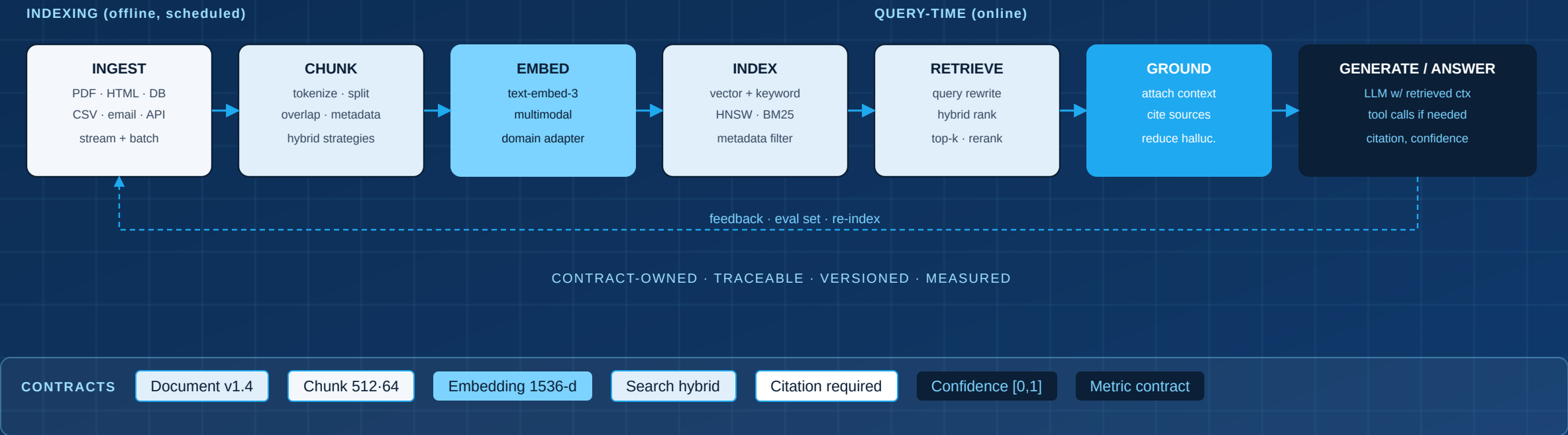
agentCard JSON · capability tags · schema version · SLA · endpoint / OIDC issuer

MESSAGE EXCHANGE

typed envelopes · idempotency-key · retries with back-off · ack / nack · streaming channels

Retrieval-Augmented Agent Pipeline

INGEST → CHUNK → EMBED → INDEX → RETRIEVE → GROUND → GENERATE

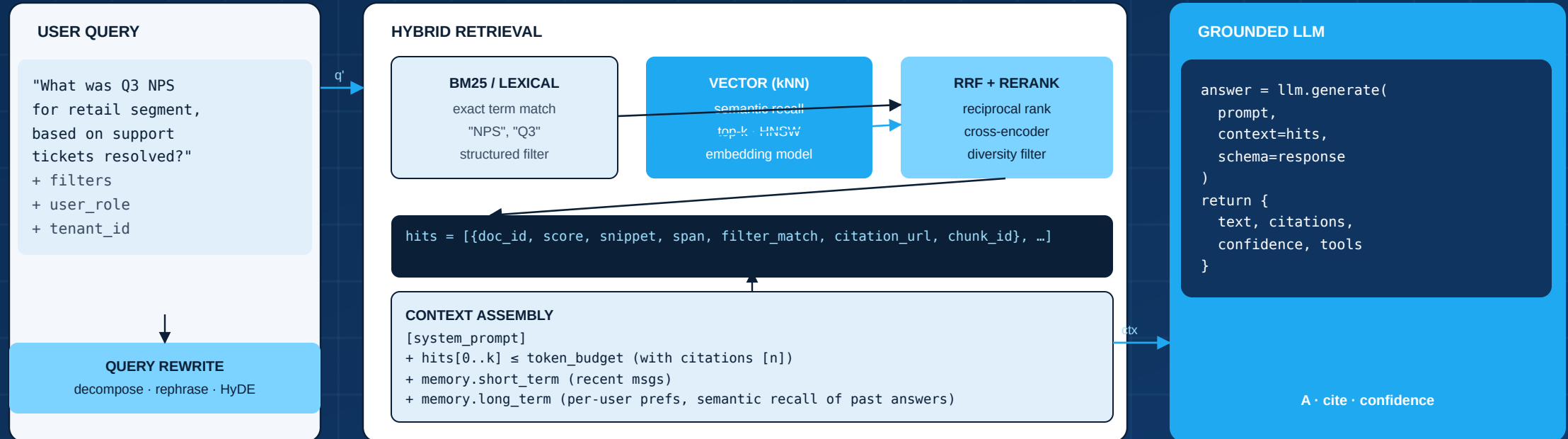


SERVICE REALIZATION · AZURE REFERENCE

STAGE	AZURE SERVICE	THROUGHPUT KNOB	QUALITY KNOB
Ingest	Event Grid · Functions · Logic Apps	event-driven triggers	parsers / OCR
Chunk + Embed	Functions · AKS · Azure OpenAI batch	batch + queue	chunk strategy
Index	AI Search (vector + BM25) · Cosmos	partition strategy	reranker model
Retrieve + Ground	AI Search retrievers · prompt flow	top-k, latency	recall + precision

Retrieval · Grounding · Memory & Context

HYBRID SEARCH · CONTEXT ASSEMBLY · CONVERSATION STATE



MEMORY LAYERS

- Short-term: rolling window of turns
- Long-term: user facts & semantic recall
- Domain knowledge: RAG index
- Working: scratchpad / tool outputs

CONTEXT WINDOW STRATEGY

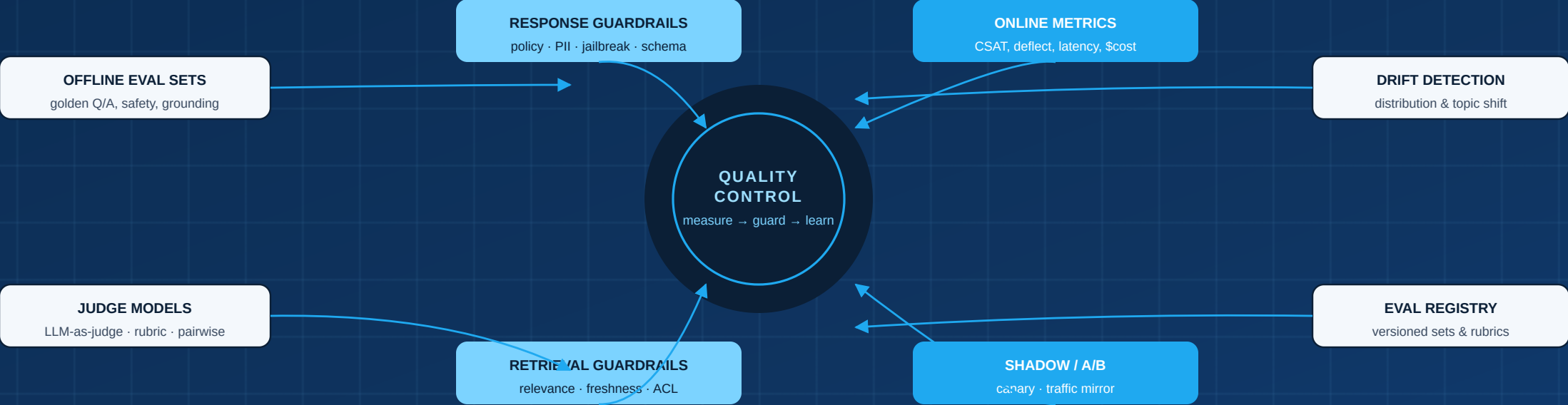
- Token budget per slot
- Truncate / summarize overflow
- Re-rank by relevance + recency
- Cite every non-system span

GROUNDING QUALITY

- Faithfulness vs. answer
- Citation coverage ≥ target
- Reject unsupported claims
- Log likelihood of context use

Evaluation, Guardrails & Continuous Quality

EVAL SETS · ONLINE METRICS · POLICY ENFORCEMENT



LAYER	OFFLINE METRIC	ONLINE METRIC	GUARDRAIL / THRESHOLD
Retrieval	Recall@k · nDCG · MRR	click-through on citations	min recall · ACL deny-list
Grounding	faithfulness, citation coverage	fact-check failure rate	reject if coverage < τ
Response	rubric score · toxicity · jailbreak	CSAT, deflection %	block + escalate on violation
Agent Loop	task success, tool-call precision	task completion rate, retries	max steps, \$cost, latency

Azure Infrastructure Blueprint for Agents

EDGE · PLATFORM · DATA · MODEL · ORCHESTRATION · CONTROL



Security · Networking · Telemetry · Cost

CROSS-CUTTING CONCERNS · POLICY · CMK · EVAL · BUDGETS

IDENTITY · SECURITY · SECRETS

Entra ID + MI

service principal
workload identity

Key Vault

CMK for embeddings
rotation · PIM

Defender for AI

prompt-injection
data exfiltration

RBAC + ABAC across agent identities · per-tool scope · deny by default
Just-in-time access for HITL approvers · break-glass account

OBSERVABILITY · EVAL · DRIFT

App Insights

trace · metrics
cost telemetry

Log Analytics

tool-call audit
LLM traces

AI Studio Eval

groundedness
safety rubric

OpenTelemetry traces · prompt & completion logging opt-in · PII redacted
SLOs: p95 latency, \$cost/conv, faithfulness, CSAT · drift alarms

NETWORKING · ISOLATION

VNet + Subnets

PE for AI Search
PE for OpenAI

Private Link

no public IP
DNS resolves inward

APIM + WAF

throttling
JWT validation

Hub-spoke topology · data exfil zones · Bastion for ops
Egress lockdown · allow-listed domains · TLS 1.3 enforced

COST · POLICY · LIFECYCLE

Cost Mgmt

\$budget per agent
alerts to owners

Azure Policy

deny public
tag mandate

Purview

data lineage
classify · DLP

PTU reservations for steady load · tokenizer pill on prompts
Retire stale agents · review & re-baseline monthly

CODE

PR · CI · SBOM

BUILD

eval gates · red team

DEPLOY

canary · blue/green

RUN

guardrails live

MEASURE

eval · drift · SLO

RETIRE

archive · replay

EVERY STAGE IS AUDITABLE · POLICY-ENFORCED · COST-CAPPED

Functional Mapping Workshop

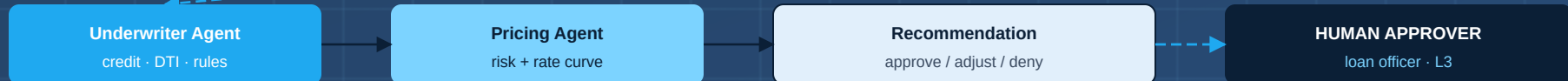
BUSINESS PROCESS → AGENT RESPONSIBILITIES → HANDOFFS

EXAMPLE · LOAN ORIGATION (L1 → L4)

L1 · INTAKE



L2 · UNDERWRITE



L3 · APPROVE



L4 · CLOSE



EVERY HANDOFF · PAYLOAD · ACK · AUDIT TRAIL

HITL Checkpoints & Exception Handling

APPROVAL · ESCALATE · CORRECT · EXCEPTION QUEUE

